FACULTY OF COMPUTING

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SECP3843 - SPECIAL TOPIC IN

DATA ENGINEERING

SECTION 01

TUTORIAL ARTICLE: DATA MODELING

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1.0 Main Problem

PT.MPM doesn't have a good system to measure and monitor the performance of its core processes. It will reduce the effectiveness of decision making from top management as their decisions are based on fragmented and manually compiled reports. This highlights the importance of centralized data to ensure competitive efficiency. Specifically:

- No real time visibility into sales KPI as example the revenue vs cost, product category performance and customer trends.
- No automated tracking of production KPIs
- Data is not integrated or visualized although the data existed across ERP, sales, inventory and maintenance systems.
- Decision making is not proactive and data driven making the process slower.

2.0 Main Solution

The solution for this problem was a Business Intelligence (BI) system with two interactive dashboards built using Google Data Studio, following the waterfall software development lifecycle (SDLC):

- Revenue, forecast accuracy, product category breakdown and monthly trend are visualized using Sales Performance Dashboard
- Product Performance dashboard will display OEE, machine-level quality/availability/performance ratios, and planned vs. actual production via gauge charts, bar charts, and heatmap pivot tables

The system then used Google Sheets as the data warehouse and will follow the ETL pipeline after alpha and beta testing with 5 PT.

Phase	Activity	Output
1. Analysis	Problem identification, literature review, data gathering	Research questions, scope definition

2. Requirement	Data analysis, system requirement analysis	Functional requirements, KPI list, user roles
3. Design	Data warehouse modelling, visualization design	DFD diagrams, approved dashboard mockups
4. Development	ETL process, data processing, dashboard building	Sales + Production BI dashboards in GDS
5. Testing	Alpha testing (8 functions), beta testing (UAT)	All 8 tests passed, 82% satisfaction score
6. Implementation	Full deployment to PT. MPM operations	Live BI dashboard in production use

Table 2.1 shows the waterfall methodology phases

3.0 User Story

User stories capture system requirements from the perspective of the people who will use the system. The following stories were derived from the actors and use cases identified in the paper's data flow diagrams and alpha test specification.

1. As a member of Top Management, I want to view sales revenue and forecast values on a single dashboard, so that I can quickly assess whether the company is meeting its sales KPIs.
2. As a member of Top Management, I want to compare planned versus actual production output per machine, so that I can evaluate the productivity of the production department.
3. As an Operations Manager, I want to monitor OEE (availability, performance, and quality) for each machine, so that I can identify which equipment is underperforming and requires intervention.
4. As a Sales Manager, I want to filter sales data by customer ID, product category, quarter, and date range, so that I can analyze sales trends for specific segments.

5. As a Data Analyst, I want to perform an ETL process that cleans and transforms raw reports into a structured warehouse, so that dashboards consistently display accurate information.
6. As an Operations User, I want to share or print the dashboard, so that I can distribute insights to stakeholders who do not have direct system access.

These user stories collectively cover both the data-input side of the system provided by Sales, Manufacturing, Inventory & Warehouse, Tech & Industrial Design, and the Data Analyst)and the data-consumption side driven by Top Management and managerial roles.

4.0 ERD Diagram

The entity Relationship Diagram (ERD) represents the entire relational database designed to support PT. MPM's Sales Order Management system. It connects raw operational shop-floor manufacturing data with the sales that feed the business intelligence dashboards.

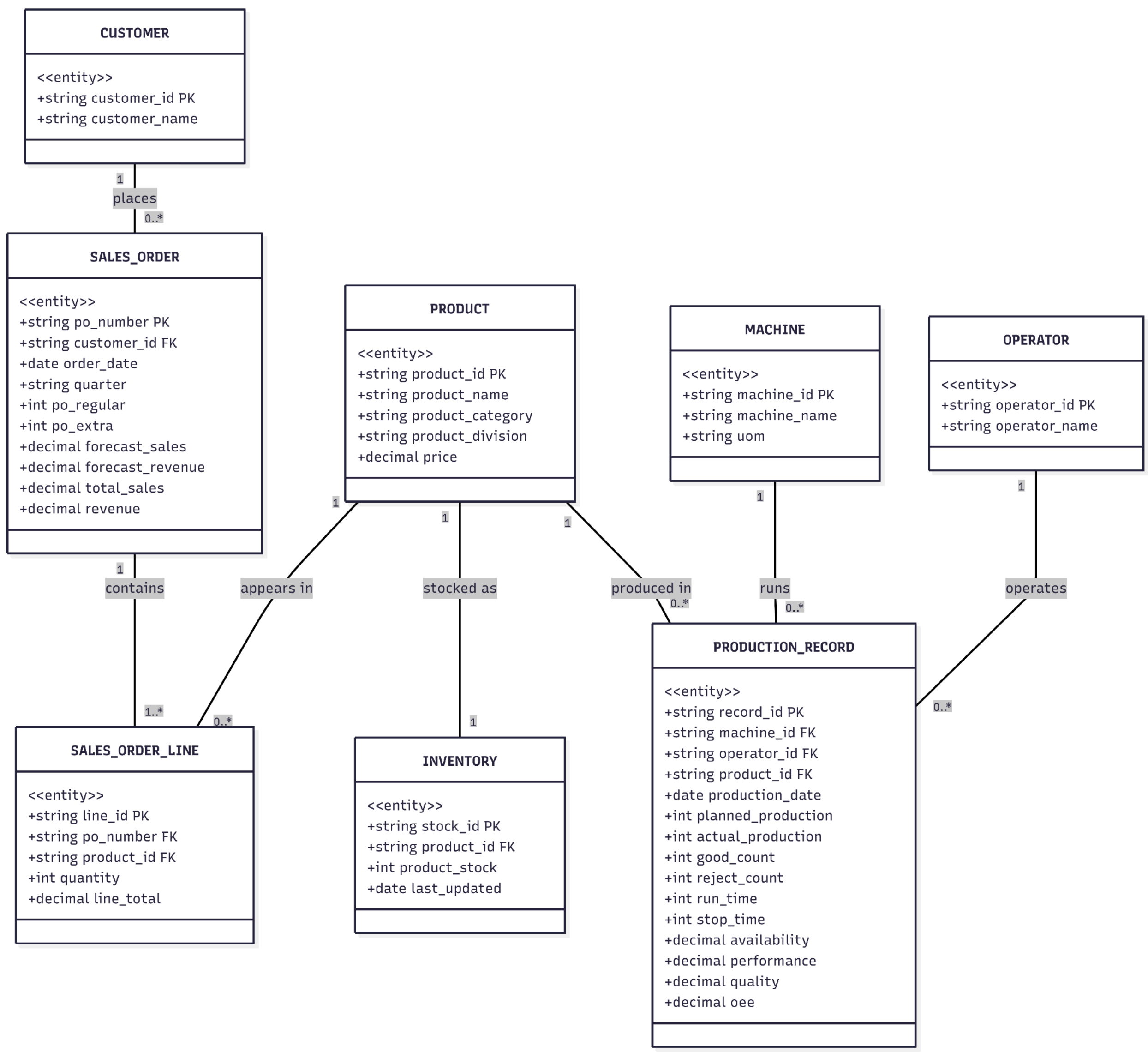


Figure 4.1: PT. MPM Sales Order Management and Production Tracking Entity-Relationship Diagram.

4.2 Data Dictionary and Schema Breakdown

The database consists of eight relational tables designed to store information with precise data types to accommodate both textual descriptions, transaction metrics, and calculated analytical decimals.

1. CUSTOMER

- customer_id (PK, string): The unique identifier for each client.
- customer_name (string): The official registered name of the business or consumer.

2. SALES_ORDER

- po_number (PK, string): The unique purchase order identifier acting as the primary key.
- customer_id (FK, string): Connects the order back to a specific row in the CUSTOMER table.
- order_date (date): The calendar date the purchase order was formally placed.
- quarter (string): Time dimension mapping used for aggregate dashboard filtering.
- po_regular (int): Count or category indicator for standard production order lines.
- po_extra (int): Count or indicator for rush/specialized custom job requests.
- forecast_sales (decimal): Pre-calculated sales projection values.
- forecast_revenue (decimal): Projected financial income based on customer sales forecasts.
- total_sales (decimal): Actual final volume or count of sales achieved.
- revenue (decimal): Confirmed monetary income generated by the specific purchase order.

3. SALES_ORDER_LINE

- line_id (PK, string): Unique row identifier for itemized breakdowns inside a purchase order.
- po_number (FK, string): Relational link mapping the line item back to its parent header in SALES_ORDER.
- product_id (FK, string): Relational link identifying which product item is being ordered.
- quantity (int): Total volume units of the product requested.
- line_total (decimal): Calculated monetary cost for the specific row

4. PRODUCT

- product_id (PK, string): Unique stock-keeping identifier for each packaging type.
- product_name (string): Descriptive label for Retail Packaging, Shipping Supplies and others.
- product_category (string): General categorization of the product.
- product_division (string): Internal business unit responsible for manufacturing the item.
- price (decimal): The standard base cost assigned per unit of the product.

5. INVENTORY

- stock_id (PK, string): Unique tracking ledger ID for stock level logs.
- product_id (FK, string): Relational link detailing which catalog product is in stock.
- product_stock (int): The exact physical count of units currently residing in the warehouse.
- last_updated (date): Timestamp logging when the stock level was last audited or modified.

6. PRODUCTION_RECORD

- record_id (PK, string): Unique identifier tracking an isolated manufacturing shift or job run.
- machine_id (FK, string): Links the production output directly to the machine asset utilized.
- operator_id (FK, string): Records the personnel member responsible for managing the run.
- product_id (FK, string): Identifies the item profile manufactured during the recorded timeframe.
- production_date (date): The calendar date the physical manufacturing took place.
- planned_production (int): Target production output volume specified by planning managers.
- actual_production (int): Total volume of items physically stamped out by the line.
- good_count (int): Total volume of manufactured items meeting quality criteria.
- reject_count (int): Defective items thrown out used to calculate structural quality ratios.
- run_time (int): Total active operating minutes of the machine.
- stop_time (int): Total downtime minutes due to maintenance, changeovers, or errors.
- availability (decimal): Ratio tracking asset uptime
- performance (decimal): Efficiency metric measuring speed deviations against optimal cycle time.
- quality (decimal): Ratio of unblemished goods produced
- oee (decimal): Overall Equipment Effectiveness

7. MACHINE

- machine_id (PK, string): Unique structural asset ID tracking hardware units on the floor.
- machine_name (string): Descriptive hardware title (e.g., Die-Cutter Line A, Slitter 02).
- uom (string): Unit of Measure standard applied to the asset's calculations.

8. OPERATOR

- operator_id (PK, string): Unique human resource identifier for shop-floor technicians.
- operator_name (string): Name of the factory employee operating the machinery.

5.0 Use Case Diagram

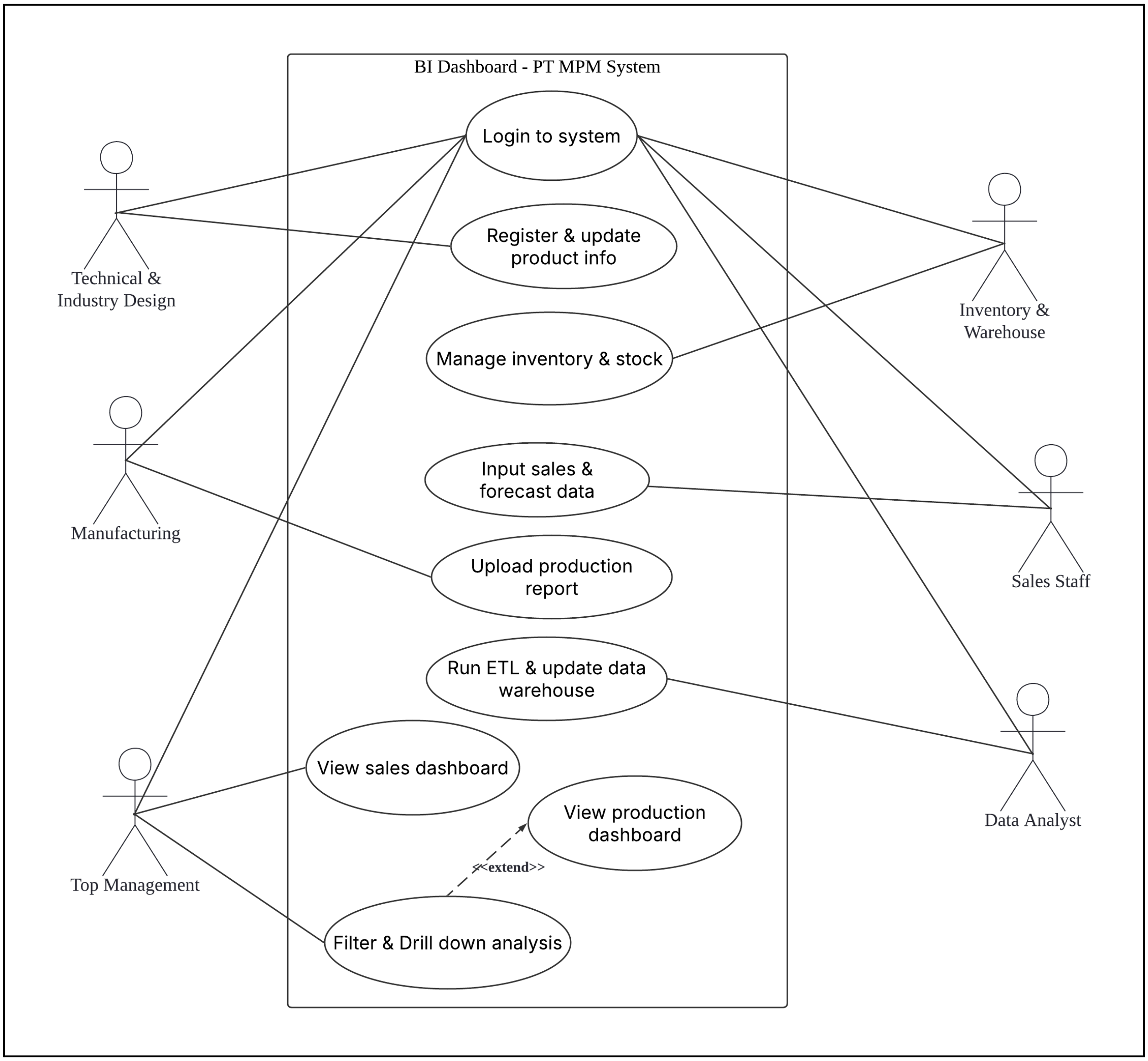


Figure 5.1 Use Case Diagram of PT MPM Dashboard System

Figure 5.1 shows the use case diagram that identifies all actors in the system and the actions they can perform. The system boundary is the BI Dashboard for PT. MPM built in Google Data Studio.

The following are the actors with the description.

1. Top Management: The primary consumer of the BI system. They view the sales performance dashboard (UC-07) and the production performance dashboard (UC-08) to

make strategic and operational decisions.

2. Technical & Industrial Design: This department registers new products and updates existing product information in the system such as product name, category, unit price, and UOM.
3. Sales Staff: Inputs sales transaction data and updates forecasts in the source system.
4. Data Analyst: Runs the ETL pipeline, cleans data, uploads to the data warehouse.
5. Inventory & Warehouse: This department manages stock levels and records product availability.
6. Manufacturing: The manufacturing department uploads production reports and maintenance logs into the system.

Table 5.1 shows use case descriptions for each with the involved actors.

#	Use Case	Actor(s)	Description
UC-01	Login to system	All actors	Every actor must authenticate via their Google account before accessing any system function.
UC-02	Register & update product info	Technical & Ind. Design	Technical staff register new product entries like name, category, unit price, and unit of measure and update these when changes occur.
UC-03	Manage inventory & stock	Inventory & Warehouse	Warehouse staff update stock levels and product availability records.
UC-04	Input sales & forecast data	Sales	Sales staff enter completed transaction data like customer ID, product sold, quantity, revenue, and forecast figures into the source system.
UC-05	Upload production report	Manufacturing	Manufacturing staff submit daily or periodic production reports and maintenance logs.
UC-06	Run ETL & update warehouse	Data Analyst	The data analyst executes the Extract-Transform-Load pipeline.
UC-07	View sales dashboard	Top Management	Top management accesses the sales performance dashboard to review KPIs.
UC-08	View production dashboard	Top Management	Top management reviews the production performance dashboard.
UC-09	Filter, drill down & export	Top Management	When viewing either dashboard, top management can apply filters.

Table 5.1 Use Case Description of PT MPM Dashboard System

6.0 Design Data/system architecture

Figure 6.1 shows the system that follows a classic 3-layer Business Intelligence architecture. Data flows from source systems through an ETL pipeline into a data warehouse, then is visualised in dashboards accessible to end users.

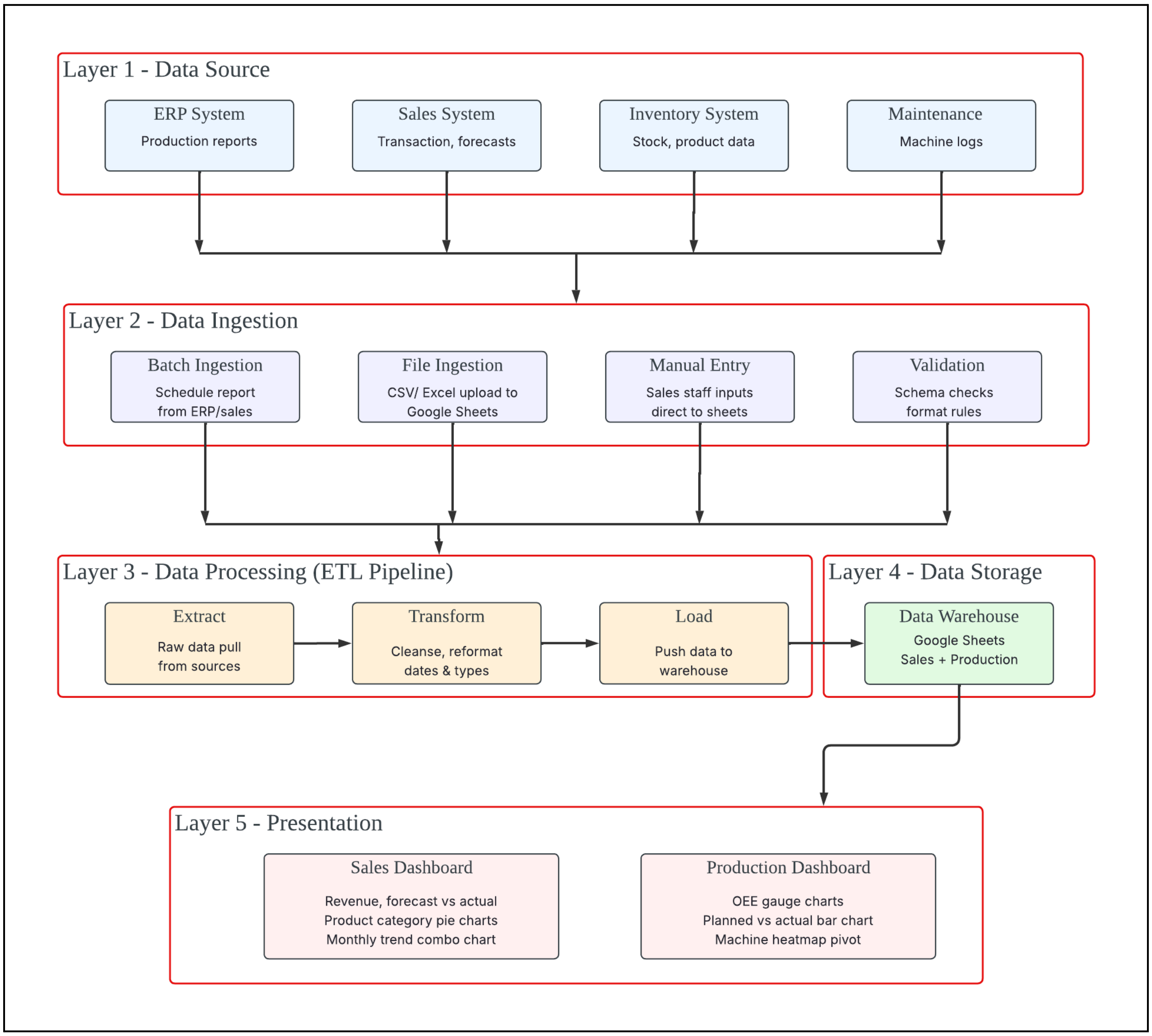


Figure 6.1 Data architecture of PT MPM Dashboard System

Layer 1 - Data Source

Four main source systems feed data into the BI pipeline:

1. ERP System: Provides production reports including machine runtime, planned production, and actual output

2. Sales System: Supplies transaction records, customer IDs, product categories, revenue, and sales forecasts
3. Inventory & Warehouse System: Maintains product stock levels and product information registered by the Technical & Industrial Design department
4. Maintenance Logs: Records machine downtime, issue types, and technician responses

Layer 2 - Data Ingestion

Data ingestion layer is a dedicated layer that sits between the source systems and the ETL pipeline and the job is to control how raw data physically enters the processing pipeline.

1. Batch ingestion: Data from the ERP and production systems is exported on a scheduled basis (daily, weekly, or at the end of each production period) as structured files.
2. File ingestion: Operations staff export reports from the ERP or accounting tools as CSV or Excel files and upload them directly into Google Sheets.
3. Manual entry: Sales staff and warehouse staff enter data directly into designated Google Sheets input tabs.
4. Validation: Sales staff and warehouse staff enter data directly into designated Google Sheets input tabs.

Layer 3 - Data Processing (ETL Pipeline)

The ETL process (Extract, Transform, Load) is the core data processing layer managed by the Data Analyst role:

1. Extract: Raw reports are pulled from the ERP and source systems and converted into Google Sheets format
2. Transform: Data cleansing removes extraneous rows; data transformation reformats fields (e.g. string dates converted to DATE type, string numbers to numeric), and table structures are split into dimension and fact components
3. Load: Cleaned, transformed data is uploaded into the Google Sheets data warehouse for use by Google Data Studio

Layer 4 - Data Storage

The data warehouse has two main table groups:

1. Sales Data Warehouse: Contains dimensions for date, customer, product, and category plus the sales fact table with revenue, forecast, and quantity measures
2. Production Data Warehouse: Contains dimensions for date, machine, operator, and product division plus the production fact table with OEE, availability, performance, quality, and planned/actual counts

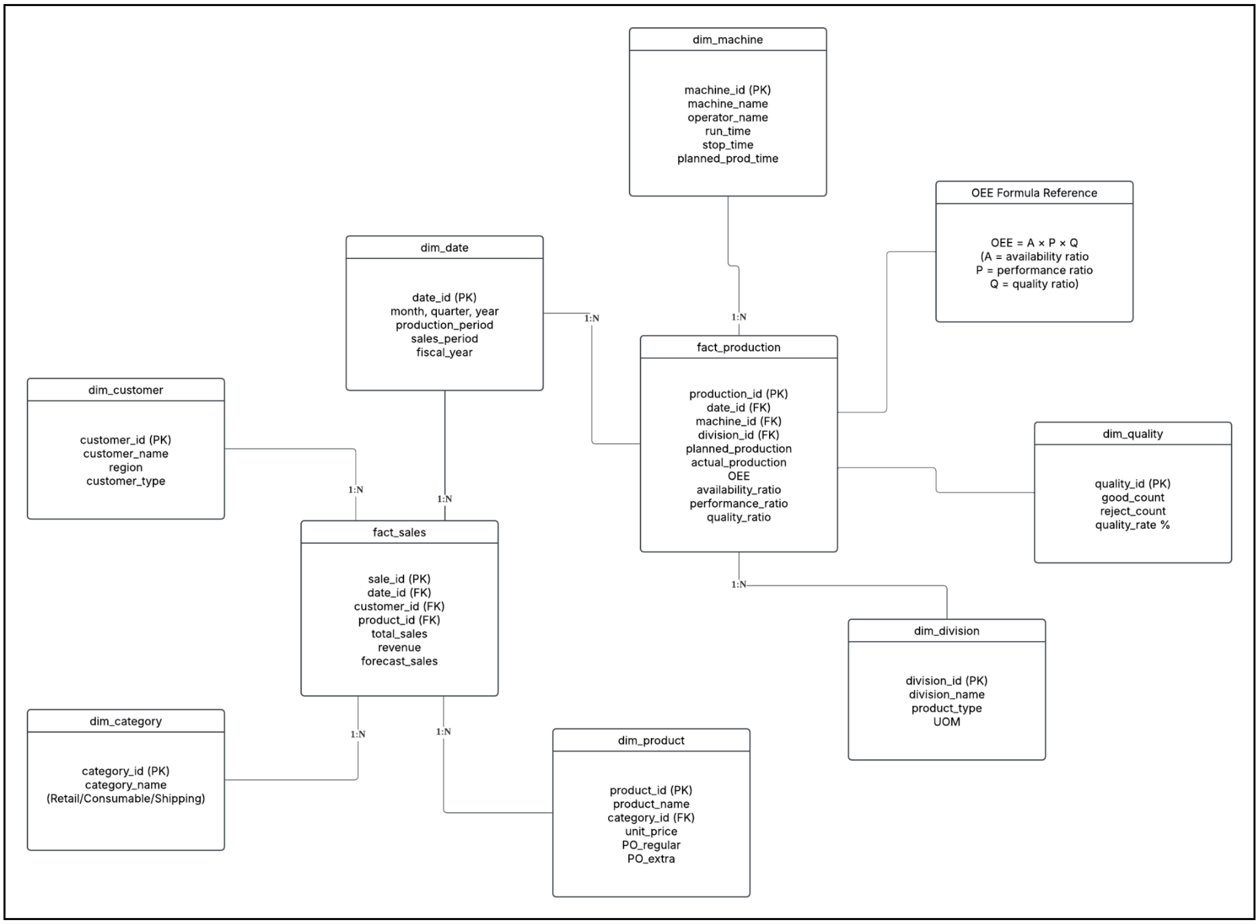
Layer 5 - Presentation (Google Data Studio)

Two dashboards are produced and combined from the data warehouse. Both dashboards are interactive web applications. Users can apply filters using drop-down menus, click chart segments to drill into subsets of data, hover over cells for exact values, and expand or collapse the pivot table using the expand-collapse function. Reports can be shared via link or exported as PDF.

7.0 Multidimensional - Galaxy Schema

7.1 Galaxy Schema PT. MPM Sales and Production

Figure 7.1 shows multidimensional analysis of sales performance across time, customer, product, and category.



7.1 Galaxy Schema PT. MPM Sales and Production

Facts Table: *fact_sales, fact_production*

- *fact_sales* table, which stores quantitative transactional measures including *total_sales*, *revenue*, and *forecast_sales*. It connects to four surrounding dimension tables via foreign keys.
- *fact_production* table holds all key operational metrics, including *planned_production, actual_production, OEE, availability_ratio, performance_ratio*, and *quality_ratio*. It connects to three dimension tables via foreign keys.

Dimension Table: *dim_customer, dim_date, dim_category, dim_product, dim_machine, dim_division*

- *dim_date*serves both fact tables by providing temporal context. For sales analysis it exposes attributes such as *month, quarter, year, sales_period,* and *fiscal_year,* while for production analysis it supports *production_date* and *production_period*, enabling time-series comparisons across both domains.
- *dim_customer* describes the buyer in each transaction, including *customer_name, region, and customer_type*, which supports segmentation by customer profile or geography.
- *dim_product* captures product-level details such as *product_name, unit_price, PO_regular, and PO_extra*, and links to *dim_category* through a category_id foreign key.
- *dim_category* classifies products into business lines enabling aggregated performance analysis by product group.
- *dim_machine* describes each piece of equipment by *machine_name, operator_name, run_time, stop_time,* and *planned_prod_time*, enabling machine-level performance benchmarking and downtime analysis.
- *dim_division* identifies the product division associated with each production run, storing *division_name, product_type,* and *UOM* (unit of measure), allowing output to be grouped by manufacturing segment.